McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Power Plant

Location ID: B196

Single-story, building which houses the central power plant for the station. Includes six 800-900kW diesel-engine generators in a high-bay generator room, a control room, offices, and a parts storage mezzanine.

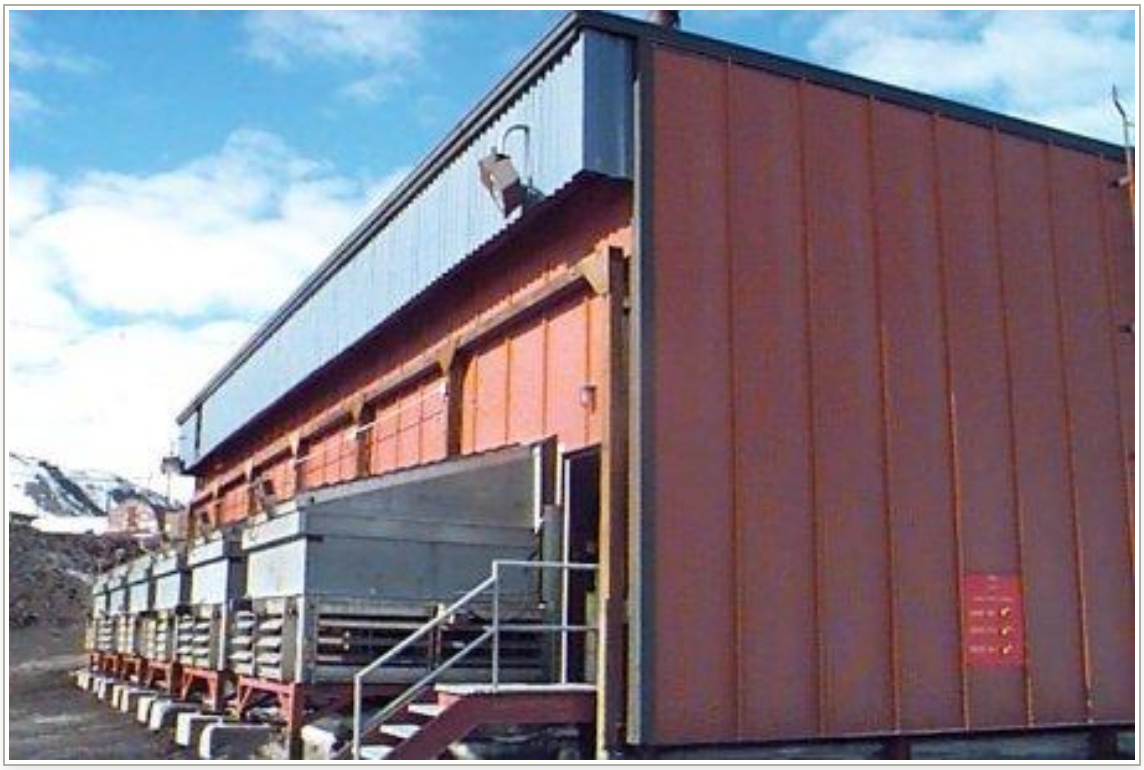


Photo 1: *Power Plant*

General Structure Information



Automatic External Defibrillator (AED) Locations

This building contains one or more AED devices that may be used in the event of life-threatening cardiac arrhythmia.

- **1EA in Generator Section**

Usage	Power Plant
Occupancy Group	B-4 Power Plant
Maximum Occupancy	19
Construction Type	Type II, 1-Hour, non-combustible; steel-framed structure with foam-insulated metal siding and roofing, steel-framed precast concrete floor, wood-framed gypsum board interior partitions.
Year Constructed	1981
Dimensions	60' x 75'10' x 25' oil storage/furnace extension

Floor Area

Floor	Area
Floor 1	4,750 sq ft
Floor 2	750 sq ft
Total	5,500 sq ft

HVAC System

Heating	One Lennox forced-air, oil-fired unit (168,000 Btu/hr)Five Chromalux wall heaters (3 at 3,000 watts; 2 at 750 watts)
Air Handling Units	None
Fuel Tank	Two 1,000 gallon day tanks and tied into Building 198's fuel tanks

Water and Sewer

Water Service	1" stub off the 4" main
Sewer Service	4" stub off the 4" main
Connections	Sinks: 2 Toilets: 1 Urinals: 0 Eyewash Station: 1 Water Cooler: 1 Two lines with three instruments
Water Heater	20 gallon, electric water heater located in the office space.

Electrical Service

Service Feeder	Feeder F
Transformer	250 KVA transformer
Electrical Service	4160/208V
Phone and LAN	Two lines with three instruments

Fire System

Fire Panel Type	Pyrotronics System 3 control panel, King-Fisher KFRTI-WP-8 radio interface/transmitter panel, pull stations, horns
Heat Detectors	Located throughout building
Suppression Type	Automatic CO2 suppression system over Cat gen-sets in main bay and in day tank fuel room. Ansul RR-1 pneumatic actuation heads are triggered by pneumatic detector on rate of temperature rise with 30-second delay upon actuation. It can be released manually at remote pull stations, and the delay has an override for immediate release. There are two 100 lb (300 lb gross) CO2 cylinders for each of six engines (12 tanks total) and two 75 lb (220 lb gross) CO2 cylinders for the fuel room.

Floorplans

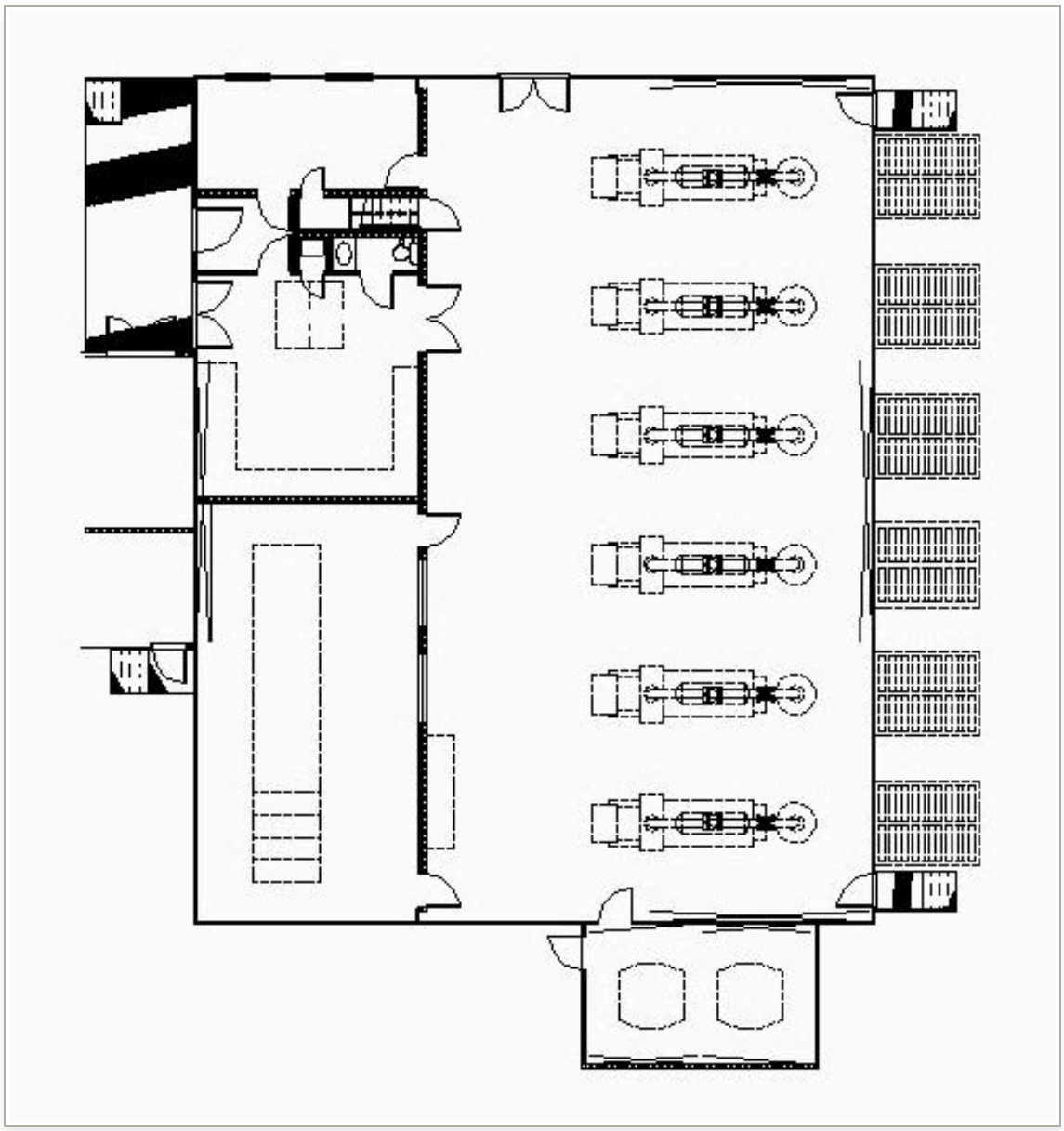


Figure 1: *Power Plant*

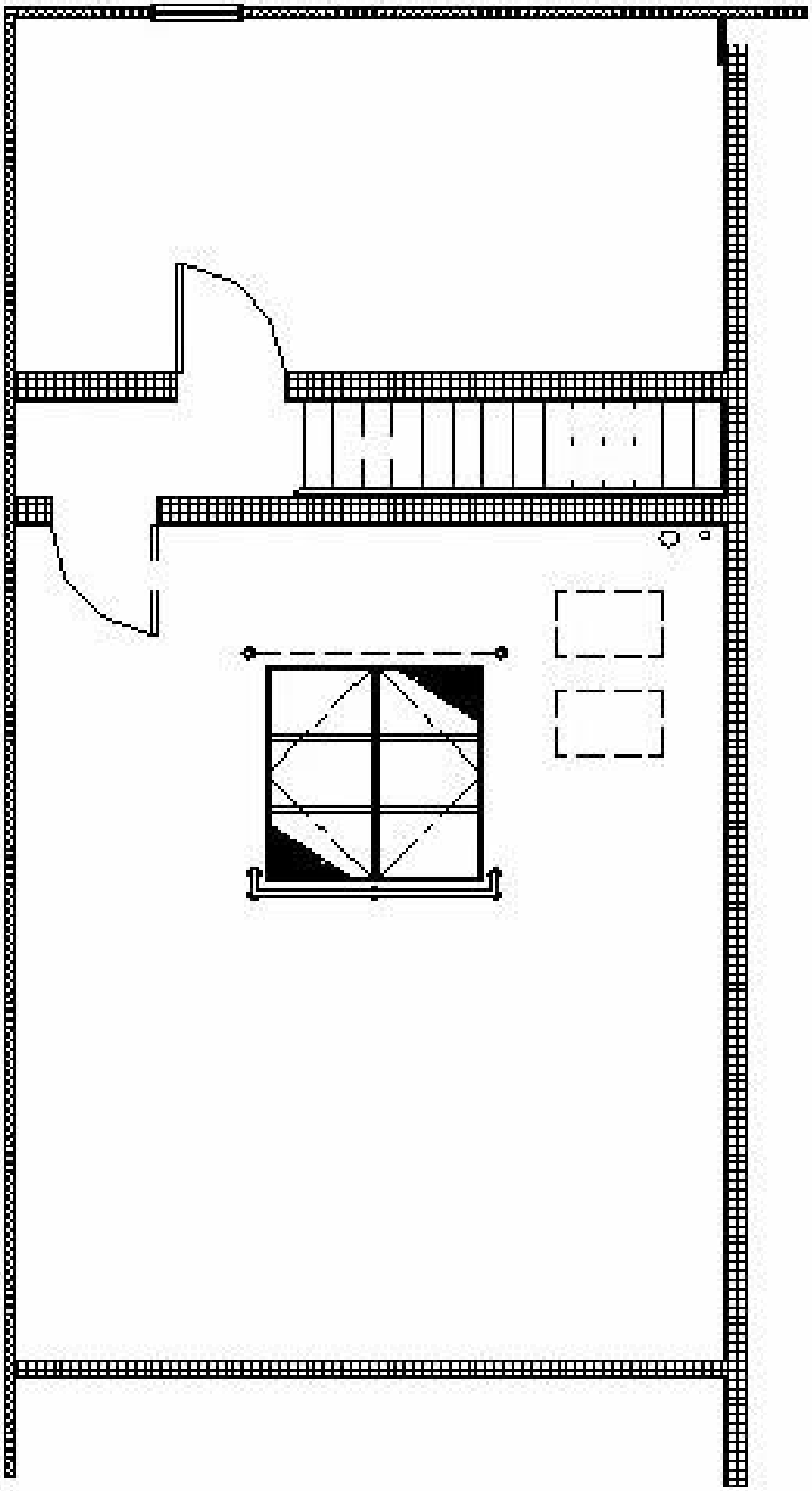


Figure 2: *Power Plant*